



**WRITING  
IT CAPSTONE  
MANUSCRIPT**

# GENERAL GUIDELINES

**PAPER SIZE: LETTER – (8.5 X 11)**

**FONT STYLE / SIZE : TIMES NEW ROMAN / 13PT**

**LINE SPACE: DOUBLE, JUSTIFIED**

**MARGINS: LEFT 1.5IN, TOP 1.5IN,  
RIGHT 1.0IN, BOTTOM 1.0IN**

**INDENTATION: 5 SPACES (NOT ONE TAB)**



# CHAPTER 1 CONTENT

01

Introduction

02

Background  
of the Study

03

Statement of  
the Problem

04

Significance  
of the Study

05

Scope and  
Limitation of the  
Study

06

Definition of  
Terms

# INTRODUCTION

- The introduction provides a general overview of the research project, highlighting its purpose, relevance, and intended contributions.
- It sets the tone for the manuscript by giving readers insight into the problem being addressed and the motivation for undertaking the study.
- This section also serves to capture interest and convey the significance of the chosen topic within the field of Information Technology.
- A well-crafted introduction positions the research within a broader context, laying the groundwork for understanding the project's objectives and approach.

# WRITING A GOOD INTRODUCTION

- Presentation of the importance of the system software. This must be the start of an introduction.
- The existence of an unsatisfactory condition; a felt problem that needs a solution. A solution that involves development of a system software.
- A desire to find a better way of doing something or improving the present system.

# BACKGROUND OF THE STUDY

- The background of the study outlines the context in which the research is conducted.
- It provides a historical and theoretical foundation for understanding the problem, detailing existing solutions and their limitations.
- This section helps establish the need for the research by demonstrating gaps in knowledge or areas for improvement.
- By reviewing relevant literature and industry practices, the background justifies the rationale for the proposed IT solution and highlights its potential contribution to technological advancements.

# STATEMENT OF THE PROBLEM

- The statement of the problem clearly defines the issue that the research aims to address.
- It articulates the specific challenges faced by stakeholders and the inadequacies of current solutions.
- This section is crucial because it frames the research questions and guides the development of objectives.
- A well-defined problem statement ensures that the study remains focused and relevant, providing a clear direction for the research process and system development.



## **GUIDELINES IN FORMULATING THE GENERAL PROBLEM AND SPECIFIC QUESTIONS**

- The opening paragraph of this part of the research paper contains the general problem of the thesis.
- The general statement of the problem and specific questions should be formulated first before developing the software programs.
- Each specific question must be clear and unambiguous.
- Answers to specific questions must contribute to the development of the whole research problem or capstone project.
- Question should start with “What”



# SIGNIFICANCE OF THE STUDY

- This section discusses the importance of the research and its potential impact on various stakeholders.
- It highlights how the project can contribute to the academic community, industry practitioners, and end users.
- The significance of the study emphasizes the value of the proposed IT solution in addressing real-world problems, enhancing operational efficiency, or improving user experiences.

## SCOPE AND LIMITATION

- The scope and limitation section defines the boundaries of the research.
- It specifies the aspects of the problem that will be addressed, the features and functionalities of the proposed solution, and the context in which the study is conducted.
- This section also acknowledges constraints, such as time, resources, and technological limitations, that may affect the project's development and implementation.
- Clearly defining the scope and limitations ensures that readers understand the focus of the research and the factors that may influence its outcomes.

# DEFINITION OF TERMS

- The definition of terms clarifies key concepts and technical jargon used throughout the manuscript.
- By providing precise definitions, this section ensures that readers, regardless of their technical background, can understand the research.
- It also establishes a common understanding of terms that may have different interpretations in other contexts.

## CHAPTER 2 CONTENT

01

Conceptual  
Literature

02

Research  
Literature

03

Technical  
Background

04

Theoretical  
Framework

05

Conceptual  
Framework

06

Synthesis

# CONCEPTUAL LITERATURE

- Introduce the key concepts, terms, and definitions related to your IT Capstone project.
- Highlight foundational works that have shaped your research area.
- Relate the concepts to your project's specific domain (e.g., software development, internet of things and data analysis).
- Make sure to cover any major research gaps or debates within this conceptual field.
- Sources or references should come from reputable and reliable sources such as books, industry reports, educational videos, or other authoritative content. Research journals and academic articles should be avoided as sources for this section.

# CONCEPTUAL LITERATURE CONTENT

Topic: *Smart Aquaculture and Automation in Fish Farming*

*Key Concepts:*

- **Smart Aquaculture:** The integration of modern technologies, such as IoT (Internet of Things), sensors, and automation, to enhance the management and monitoring of aquaculture systems (fish farms).
- **Context-Aware Systems:** A system that can sense its environment and adjust its behavior based on contextual information (e.g., water quality, temperature, fish activity).
- **Image Recognition Algorithms:** Algorithms that enable computers to identify objects and patterns in images, such as detecting fish size or water cleanliness from camera feeds.

# CONCEPTUAL LITERATURE CONTENT

Topic: Smart Aquaculture and Automation in Fish Farming

*Importance in Aquaculture:*

- The use of smart monitoring systems improves the efficiency of fish farming, optimizing feeding, water quality management, and overall fish health.
- Automation reduces the need for manual intervention, saving labor costs and ensuring more consistent care for the fish.



# RESEARCH LITERATURE

- Review previous studies and findings that are directly related to your project.
- Analyze how these studies have approached the research problem or topic.
- Show how your research will build upon or challenge previous findings.
- Discuss methodologies, results, and conclusions of relevant studies.
- Cite up-to-date, use IEEE Citation Style

# RESEARCH LITERATURE SAMPLE

Topic: Automation and Context-Aware Systems in Aquaculture

The integration of context-aware systems and image recognition algorithms into smart aquaculture has been a key area of interest in recent research, aiming to improve operational efficiency and sustainability in fish farming. [1] introduced a context-aware monitoring system designed for fish farms that uses environmental sensors (e.g., temperature, pH, and dissolved oxygen) to provide real-time data. This system adjusts essential pond operations, such as feeding and water circulation, based on the monitored environmental conditions. By implementing automated responses, the system ensures the pond environment remains optimal for fish growth while reducing resource consumption.

[1] Lee, A., B. Smith, and C. Johnson, "Context-aware smart monitoring system for fish farms using environmental sensors," Book Title/Conference/Publisher, 2020, pp. 45-56

# TECHNICAL BACKGROUND

## What to include

- Provide a deeper dive into the technical aspects of your project (tools, platforms, programming languages, technologies used).
- Explain any technical concepts or methods that are integral to understanding your research (e.g., algorithms, frameworks).
- Tailor the level of detail to the audience's expertise—whether they are technical or not.
- Visual aids (diagrams, flowcharts, etc.) can be incredibly helpful here to make complex ideas more digestible.

# TECHNICAL BACKGROUND GUIDE

Topic: Automation and Context-Aware Systems in Aquaculture

- Technologies

- **Image Recognition Algorithms**

Use **Convolutional Neural Networks (CNN)** to process live camera feeds from the pond, identifying fish behavior, detecting abnormal patterns, and monitoring water cleanliness.

**OpenCV** can be used for basic image processing tasks, while more advanced frameworks like TensorFlow can help train the image recognition models.

# TECHNICAL BACKGROUND GUIDE

Topic: Automation and Context-Aware Systems in Aquaculture

- Technologies

- **Automation of Feeding and Cleaning**

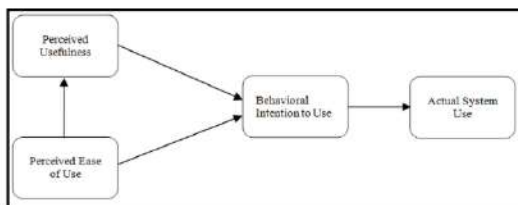
An Arduino or Raspberry Pi can be used as the central control unit for automated feeders and water circulation pumps. The system triggers feeding and cleaning based on real-time data from sensors and image recognition outputs.

# THEORETICAL FRAMEWORK

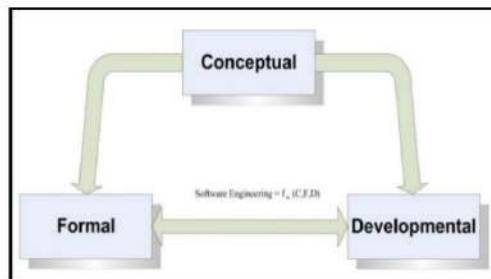
What to Include:

- Discuss the theories that underpin your research.
- Explain how these theories provide the foundation for your project's research question or hypothesis.
- Outline any models, principles, or approaches you are using to guide your study.
- Justify why these theories are the best fit for your research.

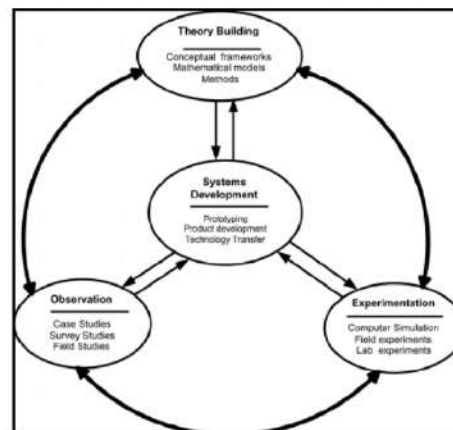
# THEORETICAL FRAMEWORK EXAMPLE



Technology acceptance mode



SERM Framework Model



Nunamaker's Multi-methodological Approach to IS Research



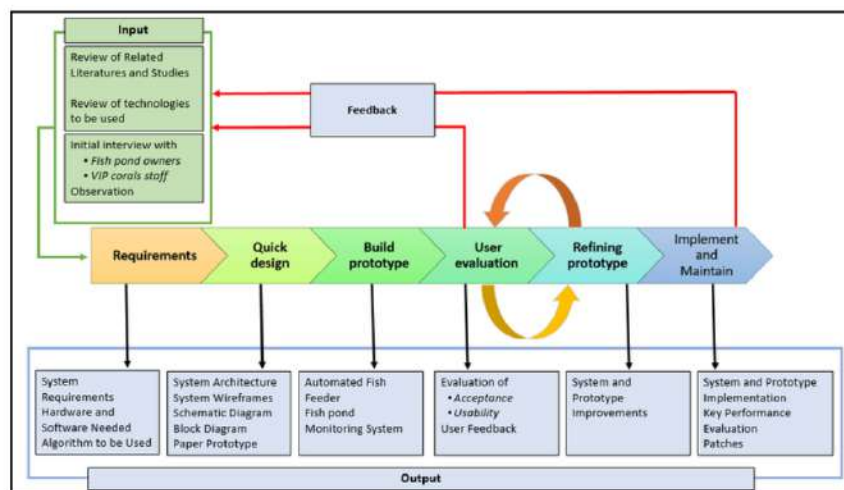
# CONCEPTUAL FRAMEWORK

## What to include

- Develop a visual model or diagram that outlines the key concepts and their relationships in your research
- Identify the key variables or components in your study.
- Explain how the conceptual framework helps guide the research design and methodology.
- This is often a visual-heavy section, so be sure the diagram is clear and easy to follow.

# CONCEPTUAL FRAMEWORK EXAMPLE

Topic: Automation and Context-Aware Systems in Aquaculture



# SYNTHESIS

## What to include

- Summarize and integrate the key insights from the literature, technical background, and theoretical/conceptual frameworks.
- Identify patterns, gaps, or contradictions within the research literature.
- Discuss how your research will contribute to filling these gaps or providing new insights.

## SYNTHESIS SAMPLE

Studies by [1] and [2] showcase how context-aware systems and image recognition technologies can independently enhance fish pond management, each focusing on different aspects of pond optimization. [1] emphasize the role of environmental sensors in creating a context-aware system that adjusts feeding schedules and water circulation based on real-time environmental data. In contrast, [2] utilize image recognition to track fish behavior, particularly movements and feeding patterns, allowing for more precise feeding schedules and reducing feed waste by responding to individual fish needs. The proposed study aims to address the gap between these two approaches by integrating both context-aware systems and image recognition technologies into a cohesive monitoring system. By combining real-time environmental monitoring with behavioral analysis, the proposed system would offer a more holistic solution to pond management, optimizing both environmental conditions and feeding efficiency simultaneously.

## CHAPTER 3 CONTENT

01

Research  
Design and  
Methods Used

02

Development  
Process

03

Programming  
Procedure

04

Hardware and  
Software  
Needed

05

Instrumentation

06

Preparation and  
Evaluation

# RESEARCH DESIGN AND METHODS USED

## Research Design

Start by explaining the overall approach to the research design of the capstone project. Discuss how the research design is structured to support the objectives of the project, including defining the research questions, hypotheses (if applicable), and expected outcomes.

Mention the research methodology you are following, such as descriptive, developmental, experimental or mixed methods, and explain how it aligns with the project goals.

# RESEARCH DESIGN AND METHODS USED

## Research Design

Discuss the methodology that will be used to gather data and evaluate the success of the project. This may include surveys, interviews, experiments, case studies, or simulations, depending on the nature of the project.

Provide a rationale for why a particular methodology was chosen and how it will help achieve the research objectives.

# RESEARCH DESIGN AND METHODS USED

How to use it:

- **System Description:** Document and describe the features, components, and operations of the developed software or system.
- **Performance Metrics:** Collect data about how well the system performs, such as speed, efficiency, user response time, and error rates.
- **User Experience (UX) Evaluation:** Gather user feedback to describe how well the system meets user needs, ease of use, and satisfaction.



# RESEARCH DESIGN AND METHODS USED

## Suggested Research Design for IT Capstone

### **Descriptive Research Design**

- This design is ideal for documenting and describing the features, functionalities, and processes of the IT system or solution being developed. It helps provide a comprehensive account of the software, hardware, or system's characteristics, behaviors, and performance.

# RESEARCH DESIGN AND METHODS USED

## Example: Research Design (Descriptive)

In this study, the researchers used the descriptive method. This method gathered the necessary information, particularly from the respondent's perspectives that is used in statistical analysis. The chosen respondents evaluated the study after it has been developed. The descriptive method was used for this study as it allows researchers to completely analyze the background of a research problem before proceeding with further investigation. It's also applied to compare various variables, as well as how different demographics react to different variables.

# RESEARCH DESIGN AND METHODS USED

## Suggested Research Design for IT Capstone

### **Developmental Research Design**

- This design is ideal when the main objective is to create, develop, or evolve a technological solution, software system, or application. Developmental research is about building or improving a product and refining it through iterative testing, feedback, and adjustments.

# RESEARCH DESIGN AND METHODS USED

How to use it:

- To use Developmental Research Design in your IT Capstone, follow this structured process that focuses on design, prototyping, testing, feedback, and refinement.

# RESEARCH DESIGN AND METHODS USED

## Example: Research Design (Developmental)

The study will utilize a Developmental Research Design to address the stated research objectives. The developmental research design is focused on the development, evaluation and refinement or enhancement of existing technologies, systems or tools. The primary goal of developmental research in IT is to develop new technologies or enhance existing ones. This involves creating prototypes, systems, or applications that address specific problems or meet particular needs. In this research a **prototype integrated with IoT devices and electrical management system will be developed and evaluated by the respondents**. The output of this proposed study will undergo various methods such as user testing, performance benchmarks, and usability evaluation. Data will be collected and analyzed and evaluate how well the technology meets the defined needs.

# RESEARCH DESIGN AND METHODS USED

How to use it:

- **Controlled Testing:** Set up controlled environments where variables (e.g., programming languages, design approaches, algorithms) can be manipulated to test their effects on the system.
- **Pre- and Post-Testing:** Use pre-tests and post-tests to evaluate changes or improvements in system performance or user engagement.
- **Hypothesis Testing:** Develop hypotheses (e.g., "Using Algorithm A will improve processing speed by 20%") and test them by comparing experimental conditions.

# RESEARCH DESIGN AND METHODS USED

## Suggested Research Design for IT Capstone

### **Experimental Research Design**

- Use this design if your project involves comparing different solutions or testing the impact of certain variables on the system. This is useful when you want to test hypotheses about system performance, algorithm effectiveness, or the impact of certain development practices.

# RESEARCH DESIGN AND METHODS USED

## Software Development Life Cycle (SDLC)

- Discuss the specific SDLC model that has been chosen for the capstone project (e.g., Agile, Waterfall, Iterative, V-Model) and explain why it aligns best with the project's goals and scope. Provide a figure for the chosen SDLC. Cite the reference where the SDLC was taken.
- Provide a discussion on the selected SDLC as the main framework to be used as guide in software development process.



# RESEARCH DESIGN AND METHODS USED

## Software Development Life Cycle (SDLC)

- Highlight how the SDLC framework ensures that the research process is organized, timely, and efficient.
- Provide a detailed discussion of each stage of the SDLC, outlining both the objectives and the methods researchers will use to conduct each phase, as well as the specific tasks they will perform during each stage. *[please refer to template for the suggested format]*

# RESEARCH DESIGN AND METHODS USED

## Software Development Life Cycle (SDLC)

### Selecting the SDLC Model:

- Discuss the specific SDLC model that has been chosen for the capstone project (e.g., Agile, Waterfall, Iterative, V-Model) and explain why it aligns best with the project's goals and scope.
- Provide a discussion on the selected SDLC as the main framework to be used as guide in software development process.
- Highlight how the SDLC framework ensures that the research process is organized, timely, and efficient.

# RESEARCH DESIGN AND METHODS USED

## Software Development Life Cycle (SDLC)

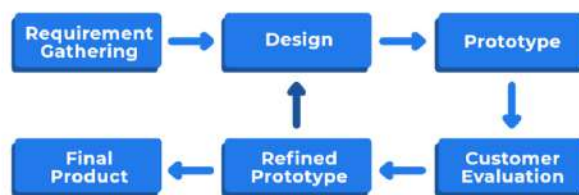


1. Cite the source / author
2. Discuss every stage

Agile Development Methodology

# RESEARCH DESIGN AND METHODS USED

## Software Development Life Cycle (SDLC)

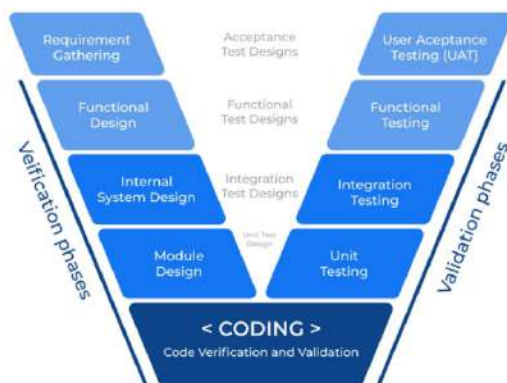


1. Cite the source / author
2. Discuss every stage

Prototyping Methodology

# RESEARCH DESIGN AND METHODS USED

## Software Development Life Cycle (SDLC)



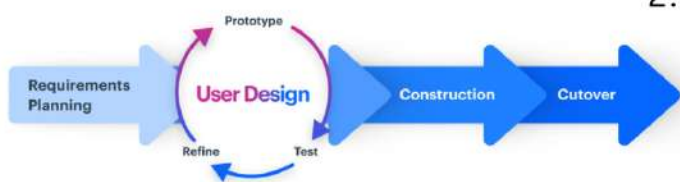
1. Cite the source / author
2. Discuss every stage

V-Model Methodology

# RESEARCH DESIGN AND METHODS USED

## Software Development Life Cycle (SDLC)

1. Cite the source / author
2. Discuss every stage



Rapid Application Development  
Methodology

# RESEARCH DESIGN AND METHODS USED

## Software Development Life Cycle (SDLC)



1. Cite the source / author
2. Discuss every stage

Spiral Methodology

# DEVELOPMENT PROCESS

- Discuss how to conduct thorough planning, gather requirements, and set clear objectives for the development phase. Emphasize the importance of aligning with project goals and user needs.
- Discuss the development process during the coding phase, including key decisions such as programming language selection, the use of libraries, and how the code is structured for scalability and maintainability.



# TESTING PROCESS

- Explain the different types of testing (unit testing, integration testing, system testing)

# TESTING PROCESS

## Testing the System/Software:

This step involves verifying that your IT solution (software, system, application, etc.) functions as expected.

- **Unit Testing:** Test individual components or modules of the software to make sure each part works correctly.
- **Integration Testing:** Check if different modules or components of the system work well together.
- **System Testing:** Test the entire system to ensure it meets requirements and performs as intended.

# TESTING PROCESS

## Testing the System/Software:

- **User Acceptance Testing (UAT):** Have users (such as your target audience or peers) test the system to see if it meets their needs and expectations.
- **Performance Testing:** Ensure that the system performs well under various conditions (e.g., handling a large volume of data or users).
- **Security Testing:** Validate the security aspects of your solution, including encryption, user authentication, and data privacy.

# VALIDATION PROCESS

- The validation process related to research in your Capstone manuscript ensures that the system you have developed is not only technically functional but also meets the intended goals, solves the problem effectively, and fulfills the research objectives. This involves evaluating how well your system satisfies the requirements defined at the beginning of the project through different validation methods.
- Benchmarking Against Existing Solutions: If your system is designed to improve upon or replace an existing solution, a common validation approach is to compare your system's performance and features against these solutions.

# DEBUGGING PROCESS

- The debugging process is essential in software development and is crucial for ensuring that your IT project works correctly and efficiently. It involves identifying, analyzing, and fixing issues or bugs in your code. Here's a structured approach to debugging.
- Debugging part tells about the methodical process of finding and reducing the number of bugs or defects in a program.

# VALIDATION PROCESS

- **Gathering User Feedback:** To validate the system from the perspective of actual users, you need to collect feedback through surveys, interviews, or direct observation. This helps evaluate if the system meets the real-world needs and expectations of users.
- **Surveys/Interviews:** Conduct interviews or surveys with a sample group of users to understand their experience with the system.
- **Usability Testing:** Evaluating the user interface (UI) and user experience (UX) is a critical step in validating whether the system meets non-functional requirements like usability and accessibility.

# PROGRAMMING PROCEDURE

- The programming procedure in your Capstone manuscript describes the process of developing the software system or solution, including the steps, techniques, tools, and technologies used to implement the system.
- This section outlines how the coding process was structured and provides insight into the development practices followed during the project. It highlights the specific tasks and decisions involved in transforming the system design and requirements into a functional, working solution.

# PROGRAMMING PROCEDURE

## System Architecture - (Figure)

- Outlines the high-level structure of the system you developed, explaining how different components interact and work together.
- It is a blueprint of the system's overall structure, including both the hardware and software elements, to ensure that the system meets its requirements.
- This should be in a form a figure



# PROGRAMMING PROCEDURE

## Discussion Contents

- **Programming Languages:** Describe the primary programming languages used in the project.
- **Frameworks and Libraries:** Specify any frameworks, libraries, or toolkits used to expedite the development process.
- **Databases and Data Storage:** If your project involves a database, mention the database management system (DBMS) used, such as MySQL, PostgreSQL, MongoDB, or SQLite, and explain the rationale for the choice.
- **Tools and Software:** List any additional tools used during development (e.g., version control, project management tools, IDEs, etc.).

# PROGRAMMING PROCEDURE

## Requirements Analysis- (figure)

- Requirements analysis encompasses those tasks that go into determining the needs or conditions to meet for a new or altered product or project, taking account of the possibly conflicting requirements of the various stakeholders, analyzing, documenting, validating and managing software or system requirements.
- This section is critical as it sets the foundation for your design and development by clearly specifying what the system must accomplish.
- The figure you include can help visualize the relationships between different requirements, such as a use case diagram, a requirements matrix, or a flowchart.

# PROGRAMMING PROCEDURE

## Use Case Diagram - (figure)

- Represents the interactions between users (or other systems) and the system being developed.
- It shows the system's functionality from an external perspective, focusing on what the system will do based on user requirements.
- A use case diagram typically includes actors (users or external systems) and the use cases (system functions) that each actor interacts with.

# PROGRAMMING PROCEDURE

## Context Diagram - (figure)

- a high-level Data Flow Diagram (DFD) that provides an overview of the system's interactions with external entities.
- It shows the system as a single process and illustrates how external actors (such as users, other systems, or external devices) interact with it.
- The context diagram helps identify the scope of the system and its boundaries, offering a simple and clear view of how data enters and exits the system.

# PROGRAMMING PROCEDURE

## Data Flow Diagram - (figure)

- is a graphical representation of the flow of data through a system, illustrating how input is transformed into output.
- It helps in visualizing how data moves through the system and how the different components of the system interact with each other. It also provides insights into the system's processes, data storage, and external entities involved.

# HARDWARE AND SOFTWARE NEEDED IN THE DEVELOPMENT

- The hardware and software requirements will depend on the nature of the system you're building.
- However, for most IT-related projects, including web, mobile, or database-driven systems, there are common hardware and software components required.
- It should in table format
- Hardware and Software Needed should be in separate table

# PROGRAMMING PROCEDURE

Entity-Relationship Diagram - (figure)

- visual representation of the data model of a system.
- It shows the entities in the system, their attributes, and the relationships between those entities.
- ERDs are commonly used in the design phase of a project to model the data that will be handled by the system.

# HARDWARE AND SOFTWARE NEEDED IN THE DEVELOPMENT

- Hardware Needed

| Component       | Description                                             | Specification                           |
|-----------------|---------------------------------------------------------|-----------------------------------------|
| Computer System | Computing machine to be used for the system development | 2.5ghz processor<br>16gb ram<br>512 SSD |
| Mobile Phone    |                                                         |                                         |
|                 |                                                         |                                         |



# HARDWARE AND SOFTWARE NEEDED IN THE DEVELOPMENT

- Software Needed

| Component        | Description                    | Specification              |
|------------------|--------------------------------|----------------------------|
| Operating System | OS for development and testing | Windows 10 Single Language |
| IDE / Editor     | For writing codes              | Visual Studio, Sublime     |
| Framework        | for building web application   | Code Igniter v.1           |

# INSTRUMENTATION

- Typically refers to the tools, equipment, or software used to collect data, monitor system performance, or evaluate results.
- The instrumentation section is essential when discussing how you measure or assess the success and functionality of your developed system.
- Discuss how will you develop an instrument such as survey form. Include the content the validation process.

# GUIDE INTERVAL FOR LIKERT SCALE QUESTIONNAIRE

- The guide interval refers to the interpretation of the numerical ratings that participants assign to the statements.
- Typically, the Likert scale measures attitudes, opinions, or perceptions, and each response point represents a level of agreement or disagreement with a statement.
- The guide interval provides a range for each rating, helping to interpret the responses more effectively.
- Include formula to get the interval

# GUIDE INTERVAL FOR LIKERT SCALE QUESTIONNAIRE

| Scale | Mean Range  | Descriptive Equivalent |                           |
|-------|-------------|------------------------|---------------------------|
| 5     | 4.21 - 5.00 | Highly Acceptable      | Strongly Agree            |
| 4     | 3.41 - 4.20 | Acceptable             | Agree                     |
| 3     | 2.61 - 3.40 | Moderately Acceptable  | Neither Agree or Disagree |
| 2     | 1.81 - 2.60 | Slightly Unacceptable  | Disagree                  |
| 1     | 1.00 - 1.80 | Not Acceptable         | Strongly Disagree         |

# GUIDE INTERVAL FOR LIKERT SCALE QUESTIONNAIRE

| Scale | Mean Range  | Descriptive Equivalent |
|-------|-------------|------------------------|
| 5     | 4.21 - 5.00 | Highly Acceptable      |
| 4     | 3.41 - 4.20 | Acceptable             |
| 3     | 2.61 - 3.40 | Moderately Acceptable  |
| 2     | 1.81 - 2.60 | Slightly Unacceptable  |
| 1     | 1.00 - 1.80 | Not Acceptable         |

# SAMPLING PROCEDURE

- The method used to select a subset of individuals or items (sample) from a larger population for the purpose of collecting data for analysis.
- It is an essential step in research methodology, as it ensures that the sample is representative of the population and helps make valid inferences about the entire population.
- Below is a guide on how to describe the sampling procedure in your Capstone manuscript.

# SAMPLING PROCEDURE

- **Sample Size Determination Formulas**

$$n = \frac{N}{1 + N(e)^2}$$

n=Sample Size  
N=Population Size  
e=Margin of error

- **Slovins Formula or YAMANE** - To make it simpler to compute the sample size without over estimating it when the population is known

$$n_0 = \frac{z^2 \cdot p \cdot (1 - p)}{e^2}$$

e: desired level of precision, the margin of error

p: the fraction of the population (as percentage) that displays the attribute

z: the z-value, extracted from a z-table<sup>62</sup>.

- **Cochran's Sample Size Formula** - Used to compute an ideal sample size for a desired level of precision, it is recommended to be used for studies with infinite populations

# SAMPLING PROCEDURE

- **Define the Population** - The first step in the sampling procedure is to define the target population, which refers to the entire group of people, items, or events that you want to study.
- **Determine Sample Size** - This step involves calculating how many participants or units should be included in the sample to ensure the results are statistically valid and representative of the population. It depends on factors like the population size, the level of confidence desired, the margin of error, and the expected variability in the population.



# SAMPLING PROCEDURE

- **Select the Sampling Method** - There are different types of sampling methods you can use, and each has its strengths and weaknesses. The choice of sampling method depends on the nature of the study and the available resources.
  - **Probability Sampling Methods (random selection)**
  - **Non-Probability Sampling Methods (non-random selection)**

# SAMPLING PROCEDURE

- **Probability Sampling Methods (random selection)**

- **Simple Random Sampling:** Each member of the population has an equal chance of being selected.
- **Systematic Sampling:** Select every k-th member from a list after choosing a random starting point.
- **Stratified Sampling:** Divide the population into strata (groups) based on a specific characteristic (e.g., age, gender, or usage level), then randomly sample from each group.
- **Cluster Sampling:** Divide the population into clusters (e.g., geographic locations), then randomly select whole clusters and sample from them.

# SAMPLING PROCEDURE

- **Non-Probability Sampling Methods (non-random selection)**

- **Convenience Sampling:** Select individuals who are easiest to reach or sample from.
- **Judgmental or Purposive Sampling:** Select individuals based on specific criteria or characteristics relevant to the study.
- **Snowball Sampling:** Participants refer others, which is often used in studies involving hard-to-reach populations.
- **Expert Sampling:** Involves selecting a sample of people who have specialized knowledge or experience about a topic.

# PARTICIPANTS OF THE STUDY

- In this section of your Capstone manuscript, you will provide a detailed information about the individuals or groups who took part in your research.
- This section is crucial because it gives context to your study, helps readers understand the demographic characteristics of your sample, and supports the validity and generalizability of your findings.

| Respondents | No. of Participants / Sample |
|-------------|------------------------------|
| IT Experts  | 10                           |
| Students    | 150                          |

# ETHICAL CONSIDERATIONS

- It is crucial to ensure that your work aligns with accepted standards, protects users and data, and follows appropriate guidelines.
- The ethical considerations in IT-related projects often focus on issues like **data privacy, security, informed consent, intellectual property**, and **fairness in algorithms or technology use**.

# ETHICAL CONSIDERATIONS

- **Informed Consent**

- In IT projects, especially those involving user data, it is essential to obtain informed consent from all participants

- **Data Privacy and Security**

- For IT projects involving user data or sensitive information, data privacy and security are primary concerns. Researchers must ensure that all collected data is securely stored and protected from unauthorized access.

# ETHICAL CONSIDERATIONS

- **Intellectual Property and Copyright**

- In IT projects, particularly those involving software development, it's critical to address intellectual property (IP) concerns.
  - **Use of Third-Party Libraries:** If you're using third-party tools, libraries, or frameworks, ensure that you have permission to use them and adhere to their licensing terms.
  - **Original Work:** Make sure your work is original, and avoid plagiarism. If you reference or build upon the work of others, you must cite your sources properly.

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# ETHICAL CONSIDERATIONS

- **Algorithmic Fairness and Bias**

- If your project involves algorithms (e.g., machine learning models, decision-making systems), you must ensure that the algorithms are fair and non-biased.
- Bias in algorithms can lead to unethical outcomes, such as discrimination against certain groups.





**END**